



An Attempt to Explain Causality in $PM_{2.5}$ and Temperature Data

A Statistical and PDE-based Approach

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Abstract

We investigate the causal influence of atmospheric temperature on $\text{PM}_{2.5}$ concentration through a dual statistical–physical framework. A Vector Autoregression with Granger causality analysis quantifies predictive influence, while a two-dimensional advection–diffusion PDE with variable wind shear and a time-dependent inversion layer models pollutant dispersion from point sources. Both approaches expose the atmospheric mixing layer as a key mediating variable in the temperature– $\text{PM}_{2.5}$ relationship.

Keywords: $\text{PM}_{2.5}$; Atmospheric Temperature; Atmospheric Mixing Layer; Granger Causality; Vector Autoregression; Advection–Diffusion Model; Inversion Layer

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1

Introduction

Repository of the work: [GitHub Repository](#)

Air pollution has emerged as one of the most significant environmental and public health challenges worldwide. Among the various atmospheric pollutants, particulate matter (PM), especially $PM_{2.5}$ and PM_{10} , poses severe risks due to its ability to penetrate the human respiratory system. According to the World Health Organization (WHO), prolonged exposure to particulate matter is associated with respiratory diseases, cardiovascular disorders, lung cancer, asthma, and premature mortality [World Health Organization, 2021](#). Fine particulate matter, particularly $PM_{2.5}$, is especially dangerous because its small size allows it to penetrate deep into the lungs and even enter the bloodstream.

Rapid urbanisation, industrial activities, vehicular emissions, and biomass burning have contributed significantly to increasing PM concentrations in many cities. In densely populated urban regions, pollutant accumulation is further influenced by meteorological variables such as temperature, humidity, wind speed, and atmospheric mixing conditions. Understanding the causal relationships between these variables and $PM_{2.5}$ concentration is therefore essential for effective forecasting and environmental policy design.

Note

This study focuses on exploring causality in $PM_{2.5}$ concentration data using statistical time series techniques. In particular, meteorological factors such as temperature are analysed to determine whether they exhibit predictive or causal influence on pollutant concentration. Such analysis is important because atmospheric conditions directly affect pollutant dispersion, chemical transformation, and accumulation processes. Improved understanding of these relationships can contribute to better forecasting systems and pollution mitigation strategies.

1.1 Types of Atmospheric Pollutants

Atmospheric pollutants are broadly classified into two categories: **primary pollutants** and **secondary pollutants**.

- **Primary pollutants** are emitted directly into the atmosphere from sources such as vehicles, industries, power plants, and biomass burning. Examples include $\text{PM}_{2.5}$, PM_{10} , carbon monoxide (CO), sulphur dioxide (SO_2), and nitrogen oxides (NO_x).
- **Secondary pollutants** are formed within the atmosphere through chemical reactions involving primary pollutants and meteorological factors such as sunlight and temperature. Examples include ozone (O_3), secondary organic aerosols, and certain fine particulate compounds.

Particulate matter itself is classified according to particle diameter:

- PM_{10} : particles with diameter $\leq 10 \mu\text{m}$,
- $\text{PM}_{2.5}$: particles with diameter $\leq 2.5 \mu\text{m}$.

PM_{10} generally affects the upper respiratory tract, whereas $\text{PM}_{2.5}$ penetrates deeper into the lungs and causes more severe health effects.

1.2 Interactions Between Atmospheric Pollutants

Atmospheric pollutants do not remain chemically isolated; instead, they interact continuously through photochemical and thermodynamic processes. Nitrogen oxides (NO_x) and volatile organic compounds (VOCs), for example, react in the presence of sunlight to produce ozone (O_3). Similarly, sulphur dioxide (SO_2) and nitrogen oxides can undergo atmospheric oxidation to form sulphates and nitrates, which contribute to secondary $\text{PM}_{2.5}$ formation.

Temperature plays a crucial role in these reactions by influencing reaction rates, atmospheric stability, and pollutant dispersion. Higher temperatures often accelerate photochemical reactions, while stable atmospheric conditions and inversion layers can trap pollutants near the surface, increasing PM concentrations.

Therefore, pollutant concentrations are governed not only by direct emissions but also by complex interactions between meteorological variables and atmospheric chemistry. This motivates the use of multivariate time series models such as Vector Autoregression (VAR) to study causal dependencies among pollutants and environmental variables.

2

Atmospheric Mixing Layer and Its Role in PM_{2.5} Dynamics

2.1 Planetary Boundary Layer

The **Planetary Boundary Layer (PBL)** is the lowest region of the atmosphere that is directly influenced by the Earth's surface through processes such as friction, heating, cooling, and moisture exchange. Its height typically ranges from a few hundred metres to several kilometres depending on meteorological conditions. Since the atmosphere within this layer responds rapidly to surface forcing, the PBL plays a central role in the transport and dispersion of atmospheric pollutants such as PM_{2.5}.

2.2 Mixing Layer

The **mixing layer** (or **mixed layer**) is the lowest portion of the **planetary boundary layer (PBL)** — the part of the troposphere directly influenced by the Earth's surface. It is the atmospheric layer near the surface where air is efficiently mixed through turbulence and convection. The principal mechanisms responsible for mixing are:

- **Thermal convection:** Solar heating warms the ground, generating buoyant air parcels that rise and mix the air column.
- **Mechanical turbulence:** Wind shear near the surface produces turbulent eddies.
- **Entrainment:** Air from the free troposphere above is gradually incorporated into the mixed layer.

The upper boundary of the mixing layer is often capped by a **temperature inversion**, where temperature increases with altitude instead of decreasing. This inversion suppresses vertical motion and limits the upward transport of pollutants and aerosols.

2.3 Diurnal Cycle of Mixing Layer Height

The mixing layer height (MLH) exhibits a pronounced daily cycle:

- **Night / Dawn:** Stable atmospheric conditions produce a shallow layer, typically

50–200 m.

- **Morning:** Solar heating initiates convection and rapid vertical mixing.
- **Afternoon:** Maximum convective activity produces deep mixing layers, often

1000–2000 m.

- **Evening:** Surface cooling suppresses turbulence and the mixing layer collapses.

At night, radiative cooling of the surface lowers the temperature of air near the ground, creating a **stable boundary layer**. Above it lies a **residual layer**, which contains remnants of the daytime mixed layer but exhibits weak turbulence.

2.4 Mathematical Framework

2.4.1 Potential Temperature

A key quantity in boundary layer meteorology is the **potential temperature**. (Stull, 1988)

$$\theta = T \left(\frac{P_0}{P} \right)^{R/c_p},$$

where:

T = absolute temperature,

P = pressure,

P_0 = 1000 hPa,

R = 287 J kg⁻¹K⁻¹,

c_p = 1004 J kg⁻¹K⁻¹.

Within a well-mixed layer, θ remains nearly constant with height. A sharp increase of θ with altitude marks the capping inversion and hence the top of the mixing layer.

Note

Parcel Method for Estimating MLH A classical graphical approach (Stull, 1988) estimates the mixing layer height as the altitude where a dry adiabatic parcel trajectory intersects the environmental temperature profile.

The dry adiabatic lapse rate is approximately

$$\Gamma_d \approx 9.8 \text{ K km}^{-1}.$$

The intersection point defines the maximum height reached by buoyant surface air parcels.

2.4.2 Convective Mixed Layer Depth

Assuming a uniform environmental lapse rate γ and cumulative surface heat flux F_H , the daytime convective mixing height satisfies (Stull, 1988)

$$z_i = \sqrt{\frac{2F_H}{\gamma}},$$

where:

z_i = mixing layer depth,

F_H = cumulative kinematic heat flux,

γ = initial lapse rate.

This relation implies:

- Larger surface heating $\Rightarrow$ deeper mixing layer.
- Stronger atmospheric stability $\Rightarrow$ shallower mixing layer.

Note

Bulk Richardson Number Criterion Under stable or nocturnal conditions, the MLH is often determined using the bulk Richardson number (Stull, 1988, Seibert et al., 2000):

$$Ri_b = \frac{(g/\theta_s)(\theta_z - \theta_s)z}{u_z^2},$$

where:

$$g = 9.81 \text{ m s}^{-2},$$

θ_s = surface potential temperature,

θ_z = potential temperature at height z ,

u_z = wind speed at height z .

When

$$Ri_b \gtrsim 0.25,$$

buoyant suppression dominates over shear-generated turbulence, signalling the top of the mixing layer.

2.5 Effect of Mixing Layer Height on PM_{2.5} Concentration

The relationship between pollutant concentration and MLH can be understood using a simple box model (Stull, 1988; Seinfeld et al., 2016):

$$C = \frac{E}{uH},$$

where:

C = pollutant concentration,

E = emission rate per unit area,

u = mean wind speed,

H = mixing layer height.

Thus,

$$C \propto \frac{1}{H}.$$

A shallow mixing layer confines pollutants within a small atmospheric volume, producing high concentrations. A deeper mixing layer dilutes pollutants over a larger volume, reducing concentration levels. (Seinfeld et al., 2016)

2.5.1 Temperature Inversion and Pollution Trapping

During inversion conditions, rising air parcels encounter warmer and therefore less buoyant air aloft (Stull, 1988; Seibert et al., 2000). Vertical motion is suppressed, and pollutants become trapped near the surface. Strong inversions may reduce MLH to 100–300 m, leading to severe pollution episodes, particularly during winter mornings and nighttime conditions.

2.6 Dependence of MLH on Temperature

Surface temperature strongly controls atmospheric mixing (Stull, 1988; Holzworth, 1964).

Table 2.1: Dependence of mixing layer height on atmospheric conditions

Condition	Effect on MLH
Strong daytime heating	Deep MLH (1000–2000 m)
Weak solar radiation	Shallow MLH
Cold nighttime surface	Stable shallow layer
Temperature inversion	Strong suppression of mixing
Summer conditions	Vigorous convection
Winter conditions	Weak mixing and stagnation

The surface heat flux approximately satisfies

$$F_H \propto (T_{\text{surface}} - T_{\text{air}}),$$

and since

$$z_i \propto \sqrt{F_H},$$

higher surface temperatures generally produce deeper mixing layers.

2.7 Motivation for Time-Series Causal Analysis

Although atmospheric boundary layer theory provides a physical explanation for the interaction between temperature and $\text{PM}_{2.5}$ concentration, the strength and direction of these relationships can vary substantially across locations, seasons, and meteorological conditions. In practice, pollutant concentrations are influenced simultaneously by emission sources, transport processes, turbulence, humidity, wind fields, chemical reactions, and boundary layer dynamics.

The mixing layer height (MLH) plays a particularly important role because it determines the effective vertical volume available for pollutant dispersion. A shallow mixing layer restricts vertical transport and promotes pollutant accumulation, whereas a deeper convective layer enhances dilution and turbulent mixing. Consequently, MLH strongly modulates the observed relationship between temperature and particulate concentration (Seibert et al., 2000; Dang et al., 2019; Y. Miao et al., 2019).

However, physical mechanisms alone do not determine whether statistically significant temporal dependencies are observable in atmospheric monitoring data. For this reason, Vector Autoregressive (VAR) models and Granger causality analysis are useful tools for studying the lagged interdependence between meteorological variables and pollutant concentrations. These methods help identify whether past values of temperature contain predictive information about future $PM_{2.5}$ concentrations (and vice versa), after accounting for the temporal structure of the multivariate system.

Thus, Granger causality analysis should not be interpreted as establishing true physical causation; rather, it provides a statistical framework for analysing directional temporal dependence and dynamic interactions within atmospheric systems.

3

Problem Statement

The primary objective of this study is to investigate the dynamic relationship between atmospheric temperature and $\text{PM}_{2.5}$ concentration, with particular emphasis on determining whether temperature variations possess predictive power for particulate pollution levels.

Air pollution is governed by a combination of emission processes, atmospheric transport, turbulence, chemical transformation, and meteorological forcing. Among meteorological variables, temperature plays a particularly important role because it directly influences atmospheric stability, boundary layer development, turbulence intensity, and pollutant dispersion. Consequently, understanding the causal interaction between temperature and $\text{PM}_{2.5}$ is essential for air quality forecasting and environmental policy analysis. The problem is approached in two stages:

1. **Statistical causality analysis using Vector Autoregressive (VAR) models and Granger causality testing.**
2. **Physical modelling of pollutant dispersion using a point-source diffusion framework for $\text{PM}_{2.5}$ forecasting under varying temperature conditions.**

4

Data Description

4.1 Air Quality Monitoring Station Data

Air Quality Monitoring Station (AQMS) data were provided by the West Bengal Pollution Control Board from 307 AQMS stations located across West Bengal. From each AQMS station, six variables were observed: temperature (measured in degree Celsius), relative humidity (measured in percentage), SO₂ concentration, NO₂ concentration, PM_{2.5} concentration, and PM₁₀ concentration, with the pollutant concentrations measured in $\mu\text{g}/\text{m}^3$. The six variables were encoded as shown in Table 2. The available data consisted of hourly averages of these six variables for each day from 01.07.2025 to 29.10.2025. AQMS stations continuously monitor ambient air quality using automated sensors and analyzers installed at fixed monitoring locations. Hourly observations represent hourly averaged concentrations of pollutants and meteorological variables, computed from measurements recorded at finer temporal resolutions.

Table 4.1: *Variable Encoding in AQMS data*

Variable	Encoding
ext_temp	Temperature
ext_humi	Relative Humidity
so2	SO ₂
no2	NO ₂
pm10	PM ₁₀
pm25	PM _{2.5}

4.2 Manual Station Data

The primary dataset, comprising hourly Particulate Matter (PM) concentrations and aggregated Air Quality Indices (AQI), was systematically harvested from the Central Pollution Control Board (CPCB) [live air quality dashboard](#), which hosts the localized

monitoring data for the West Bengal Pollution Control Board (WBPCB) network. Because the portal operates as a Single Page Application (SPA) devoid of a public-facing developer API, a custom programmatic extraction pipeline was engineered to capture the data.

Data Pipeline Workflow

Network traffic analysis was initially employed to reverse-engineer the frontend interactions and isolate the underlying undocumented JSON API endpoints responsible for data delivery. To facilitate automated retrieval across the distributed network of monitoring stations, a multithreaded Python architecture utilizing concurrent execution pools was implemented. This allowed for parallel asynchronous requests, significantly mitigating the high I/O-bound latency inherent to sequential web scraping.

Stringent session management protocols were enforced throughout the extraction process. Explicit HTTP headers—specifically configuring the `User-Agent`, `Referer`, and `Origin`—were injected into the payloads to emulate legitimate browser environment requests and successfully bypass server-side WAF (Web Application Firewall) bot mitigation strategies. Furthermore, robust exception-handling heuristics were integrated to gracefully manage socket timeouts and JSON decode anomalies, which are typical artifacts of server-side rate-limiting or payload obfuscation.

Geospatial Coordinate Resolution and Meteorological Integration

To construct a comprehensive spatiotemporal model, the integration of precise meteorological covariates like windspeed and direction data were essential, specifically for solving the PDE-based model of the flow of particulate matters.

Data Pipeline Workflow

First, the unstructured, nominal location strings of the WBPCB monitoring stations were converted into deterministic geospatial coordinates (latitude and longitude) via a programmatic geocoding pipeline. This spatial resolution was achieved by interfacing with the Ola Maps Geocoding REST API. Structured HTTP GET requests, enriched with requisite correlation headers and URL-encoded query parameters, were systematically dispatched. The JSON responses were parsed to isolate the highest-confidence geometric matches for each station, establishing a rigid spatial anchor for the dataset.

Upon acquiring these precise geolocations, spatiotemporally synchronized atmospheric data was integrated using the Open-Meteo historical API. A subsequent extraction script dynamically queried the Open-Meteo endpoints utilizing the newly resolved coordinate pairs. This facilitated the localized extraction of critical meteorological parameters—such as wind direction, and wind velocity—at an hourly temporal

resolution matching the pollution dataset. The final synthesized data structure deterministically aligns the geocoded meteorological variables with the primary pollutant concentrations, yielding a high-fidelity dataset suitable for advanced downstream modeling.

5

Main Workflow

The main workflow is motivated by the work of (Ping et al., 2025). For certain AQMS stations, one or both of the variables `so2` and `no2` contained more than 20% missing observations relative to the total number of rows. An attempt was made to cross-reference the missing values using other sources, such as the CPCB Air Quality Portal (the website of India's Central Pollution Control Board) and `data.gov.in`, but with limited success. Therefore, for all such stations, the workflow outlined below was implemented after excluding the columns `so2` and `no2`.

For AQMS stations where missing data were present but constituted less than 20% of the total number of rows, the rows containing missing values were removed, and the workflow was subsequently carried out.

Algorithm 1 Workflow for Stationarity Analysis, VAR Modelling, Granger Causality, FEVD and IRF Analysis

1: **Input:** Time series variables

$\{\text{ext_temp}, \text{ext_humi}, \text{pm25}, \text{pm10}, \text{so2}, \text{no2}\}$

2: Fix significance level $\alpha = 0.05$

Step 1: Stationarity Checking

3: **for** each variable X in the set of six time series **do**

4: Initialise differencing order: $d \leftarrow 0$

5: **repeat**

Step 1.1: Selection of ADF Lag Order

6: **for** $h = 1$ to 400 **do**

7: Fit the ADF regression model for X using lag h

8: Extract residuals: $\hat{\varepsilon}_t^{(h)}$

9: Perform Ljung-Box test on $\hat{\varepsilon}_t^{(h)}$

10: **if** Ljung-Box test accepts white-noise hypothesis at level α **then**

11: Select optimal lag: $h^* \leftarrow h$

12: **break**

13: **end if**

14: **end for**

Step 1.2: ADF Testing

15: Fit the ADF regression model using lag h^*

16: Perform Augmented Dickey-Fuller (ADF) test on X

Step 1.3: KPSS Validation

17: Perform KPSS test on X

18: **if** ADF test rejects the unit-root null hypothesis **then**

and KPSS test accepts the stationarity null hypothesis

19: Mark series X as stationary

20: **else**

21: Difference the series: $X \leftarrow \nabla X$

22: Update differencing order: $d \leftarrow d + 1$

23: **end if**

24: **until** X is stationary

25: **end for**

Step 2: Pairwise VAR Modelling and Granger Causality

26: **for** every pair of stationary variables (X_i, X_j) **do**

27: Fit a bivariate VAR model:

$\text{VAR}(X_i, X_j)$

28: Perform Granger causality tests:

$X_i \Rightarrow X_j \quad \text{and} \quad X_j \Rightarrow X_i$

29: **end for**

30: **Output:**

- Stationary transformed series
 - Pairwise Granger causality relationships
-

6

An Overview of Time Series Methods Used

6.1 Augmented Dickey-Fuller Test

The Augmented Dickey-Fuller test was introduced by Said and Dickey in 1984 in (Said et al., 1984). It involves testing for a unit root in the model,

$$\Delta X_t = \alpha + \beta t + \gamma X_{t-1} + \sum_{i=1}^p \delta_i \Delta X_{t-i} + \varepsilon_t$$

where we assume that $\{\varepsilon_t\}$ is white noise. ADF tests the following :

$$H_0 : \gamma = 0, \text{ vs } H_1 : \gamma < 0$$

Under the null, the time series is a random walk which wanders off to infinity. Under the alternate, we are able to show that it is asymptotically weakly stationary.

Consider the simplified ADF model

$$X_t = \rho X_{t-1} + \varepsilon_t,$$

where $\{\varepsilon_t\}$ is a white-noise process satisfying

$$E(\varepsilon_t) = 0, \quad \text{Var}(\varepsilon_t) = \sigma^2, \quad \text{Cov}(\varepsilon_t, \varepsilon_{t-h}) = 0 \quad (h \neq 0).$$

Under the alternative hypothesis of the ADF test,

$$|\rho| < 1.$$

We now show that the process is weakly stationary by expressing it as an infinite moving-average process.

6.1.1 Moving Average Representation

Repeated substitution gives

$$\begin{aligned}
 X_t &= \rho X_{t-1} + \varepsilon_t \\
 &= \rho(\rho X_{t-2} + \varepsilon_{t-1}) + \varepsilon_t \\
 &= \rho^2 X_{t-2} + \rho \varepsilon_{t-1} + \varepsilon_t \\
 &\vdots \\
 &= \rho^k X_{t-k} + \sum_{j=0}^{k-1} \rho^j \varepsilon_{t-j}.
 \end{aligned}$$

Since $|\rho| < 1$,

$$\rho^k \rightarrow 0 \quad \text{as } k \rightarrow \infty.$$

Hence,

$$X_t = \sum_{j=0}^{\infty} \rho^j \varepsilon_{t-j}.$$

Thus X_t admits the infinite moving average representation

$$X_t = \sum_{j=0}^{\infty} \rho^j \varepsilon_{t-j}.$$

Mean

Using linearity of expectation,

$$E(X_t) = E\left(\sum_{j=0}^{\infty} \rho^j \varepsilon_{t-j}\right).$$

Therefore,

$$E(X_t) = \sum_{j=0}^{\infty} \rho^j E(\varepsilon_{t-j}).$$

Since $E(\varepsilon_t) = 0$,

$$E(X_t) = 0.$$

Variance

Using the fact that white-noise terms are uncorrelated,

$$\text{Var}(X_t) = \text{Var}\left(\sum_{j=0}^{\infty} \rho^j \varepsilon_{t-j}\right).$$

Hence,

$$\text{Var}(X_t) = \sum_{j=0}^{\infty} \rho^{2j} \text{Var}(\varepsilon_{t-j}).$$

Since $\text{Var}(\varepsilon_t) = \sigma^2$,

$$\text{Var}(X_t) = \sigma^2 \sum_{j=0}^{\infty} \rho^{2j}.$$

The geometric series converges because $|\rho| < 1$:

$$\sum_{j=0}^{\infty} \rho^{2j} = \frac{1}{1 - \rho^2}.$$

Therefore,

$$\boxed{\text{Var}(X_t) = \frac{\sigma^2}{1 - \rho^2}}.$$

Autocovariance Function

For $h > 0$,

$$\gamma(h) = \text{Cov}(X_t, X_{t-h}).$$

Using the Moving Average representation,

$$X_t = \sum_{j=0}^{\infty} \rho^j \varepsilon_{t-j},$$

and

$$X_{t-h} = \sum_{k=0}^{\infty} \rho^k \varepsilon_{t-h-k}.$$

Hence,

$$\gamma(h) = \text{Cov}\left(\sum_{j=0}^{\infty} \rho^j \varepsilon_{t-j}, \sum_{k=0}^{\infty} \rho^k \varepsilon_{t-h-k}\right).$$

Using the fact that the white-noise terms are uncorrelated,

$$\gamma(h) = \sigma^2 \sum_{j=0}^{\infty} \rho^j \rho^{j+h}.$$

Thus,

$$\gamma(h) = \sigma^2 \rho^h \sum_{j=0}^{\infty} \rho^{2j}.$$

Therefore,

$$\gamma(h) = \frac{\sigma^2}{1 - \rho^2} \rho^h.$$

6.1.2 Conclusion

We have shown that

$$E(X_t) = 0,$$

$$\text{Var}(X_t) = \frac{\sigma^2}{1 - \rho^2},$$

and

$$\gamma(h) = \frac{\sigma^2}{1 - \rho^2} \rho^h,$$

which depends only on the lag h and not on time t . Hence the process is weakly stationary whenever

$$|\rho| < 1.$$

Therefore, under the alternative hypothesis of the ADF test, the process is asymptotically weakly stationary.

6.1.3 Generalisation to the Augmented Dickey–Fuller Regression

The same argument extends to the general ADF regression

$$\Delta X_t = \alpha + \beta t + \gamma X_{t-1} + \sum_{i=1}^p \delta_i \Delta X_{t-i} + \varepsilon_t.$$

Rewriting in level form yields an autoregressive process of order $p + 1$:

$$X_t = c + \beta t + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \phi_{p+1} X_{t-p-1} + \varepsilon_t,$$

where the coefficients ϕ_i are determined by γ and the augmentation parameters δ_i . If the characteristic polynomial

$$\Phi(z) = 1 - \phi_1 z - \phi_2 z^2 - \cdots - \phi_{p+1} z^{p+1}$$

has all its roots outside the unit circle, then the process admits a convergent infinite moving average representation of the form

$$X_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j},$$

with absolutely summable coefficients satisfying

$$\sum_{j=0}^{\infty} |\psi_j| < \infty.$$

Consequently,

$$E(X_t), \quad \text{Var}(X_t), \quad \text{and} \quad \gamma(h) = \text{Cov}(X_t, X_{t-h})$$

exist and are finite, with $\gamma(h)$ depending only on the lag h . Thus, after removing any deterministic trend component βt , the ADF process is asymptotically weakly stationary under the alternative hypothesis.

6.2 KPSS Test

The Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test (Kwiatkowski et al., 1992) is a statistical test used to determine whether a time series is stationary around a level or deterministic trend. Unlike the Augmented Dickey–Fuller (ADF) test, where the null hypothesis assumes the presence of a unit root, the KPSS test takes stationarity as the null hypothesis.

For a time series $\{X_t\}$, the hypotheses of the KPSS test are

Hypothesis

$$H_0 : \text{The series is stationary}$$

against

$$H_1 : \text{The series contains a unit root and is non-stationary.}$$

The test statistic is based on the partial sums of residuals obtained after regressing the series on a constant or on a constant and deterministic trend. Let the residuals be denoted by $\hat{\varepsilon}_t$, and define the cumulative residual process

$$S_t = \sum_{i=1}^t \hat{\varepsilon}_i.$$

The KPSS statistic is given by

$$\text{KPSS} = \frac{1}{n^2 \hat{\sigma}^2} \sum_{t=1}^n S_t^2,$$

where $\hat{\sigma}^2$ is an estimator of the long-run variance of the residual process. Large values of the test statistic provide evidence against the null hypothesis of stationarity.

The KPSS test is commonly used alongside the ADF test, since the two tests have complementary null hypotheses and together provide stronger evidence regarding the stationarity properties of a time series.

6.3 Ljung-Box Test

The Ljung-Box test was introduced by Ljung and Box in [Ljung et al., 1978](#). The Ljung-Box test is a statistical test used to determine whether a time series (or the residuals of a fitted model) exhibits significant autocorrelation. The test statistic is

$$Q = n(n+2) \sum_{k=1}^h \frac{\hat{\rho}_k^2}{n-k}$$

which asymptotically follows χ^2 distribution with h degrees of freedom under the null (i.e. series is white noise).

6.4 Vector Autoregressive (VAR) Model

A Vector Autoregressive (VAR) model is a multivariate extension of the autoregressive (AR) model used to capture linear interdependence among multiple time series. In a VAR model, each variable is expressed as a linear function of its own past values as well as the past values of all other variables in the system.

A VAR(p) model with k variables is given by

$$\mathbf{Y}_t = \mathbf{c} + A_1 \mathbf{Y}_{t-1} + A_2 \mathbf{Y}_{t-2} + \cdots + A_p \mathbf{Y}_{t-p} + \boldsymbol{\varepsilon}_t,$$

where $\mathbf{Y}_t$ is a $k \times 1$ vector of endogenous variables, $\mathbf{c}$ is a vector of constants, A_i are coefficient matrices, and $\boldsymbol{\varepsilon}_t$ is a vector white-noise process. VAR models are widely used in economics, finance, meteorology, and environmental studies for analysing dynamic relationships among variables.

6.5 Granger Causality

Granger causality is a statistical concept used to determine whether one time series contains predictive information about another. A variable X_t is said to “Granger-cause”

another variable Y_t if past values of X_t help improve the prediction of Y_t beyond the information contained in past values of Y_t alone.

Within the VAR framework, Granger causality is tested by examining whether the lagged coefficients of one variable are jointly significant in the equation of another variable. The null hypothesis is generally formulated as

$$H_0 : \text{Lagged values of } X_t \text{ do not affect } Y_t.$$

This is usually tested using an F -test or Wald test. Rejection of the null hypothesis suggests the presence of predictive causality between the variables.

7

Results and Discussion

7.1 Granger Causality Diagrams

Through schematic diagrams, we illustrate Granger causality relationships between pairs of variables using directed graphs. Each directed edge represents significant Granger causality at the 5% significance level. Edges drawn with a bold solid line indicate that the p-value associated with the Granger causality test between the corresponding pair of variables (represented by the vertices of the graph) is less than 0.05. A dashed edge indicates that the corresponding p-value lies between 0.05 and 0.10. The direction $X \rightarrow Y$ signifies that variable X Granger-causes variable Y .

Due to paucity of space, all of the 307 Granger causality diagrams corresponding to the 307 AQMS stations from which data was collected, cannot be included in the report. Instead, they are made available at the repository, specified tabularly in the [Appendix A](#).

7.2 Summary of Results

We observe the following types of results in the preceding table:

1. Temperature Granger causes both PM_{10} and $PM_{2.5}$, and vice-versa.
2. Temperature Granger causes both PM_{10} and $PM_{2.5}$, and only one of PM_{10} or $PM_{2.5}$ Granger cause temperature.
3. Temperature Granger causes both PM_{10} and $PM_{2.5}$, but none of them Granger cause temperature.
4. Temperature Granger causes exactly one of PM_{10} and $PM_{2.5}$, and that variable also Granger causes temperature while the other variables does not.
5. Temperature Granger causes neither PM_{10} nor $PM_{2.5}$, yet both of them Granger cause temperature.
6. Temperature Granger causes neither PM_{10} nor $PM_{2.5}$, yet exactly one of them Granger cause temperature.

7. No Granger causality between temperature and $PM_{2.5}$ or PM_{10} .

The above results indicate that the relationship between temperature and particulate matter concentrations is highly heterogeneous across different locations. In several cases, bidirectional Granger causality is observed, suggesting a feedback relationship between temperature and particulate pollutants. In other cases, the causality is unidirectional, indicating that either temperature influences particulate concentration dynamics or particulate matter contributes to variations in temperature. There also exist locations where no statistically significant Granger causal relationship is detected. Overall, these findings suggest that the interaction between meteorological variables and air pollutants is strongly location-dependent and may be influenced by local environmental, climatic, and anthropogenic factors.

8

Limitations of the VAR Model

Although the VAR framework enables us to study predictive relationships and temporal interactions among the AQMS variables, the present approach has several important limitations:

1. The model does not explicitly account for spatial dependence among monitoring stations. Geographically nearby locations are likely to exhibit similar pollution characteristics due to the transport and dispersion of pollutants such as $\text{PM}_{2.5}$ and PM_{10} .
2. VAR models primarily capture linear interdependencies among variables. However, atmospheric and environmental processes are often highly non-linear, which may limit the model's ability to fully describe complex pollutant dynamics.
3. Important meteorological variables such as wind speed and wind direction have not been incorporated into the present analysis, due to the lack of data. Since wind plays a crucial role in the transport, dispersion, and accumulation of pollutants, excluding these variables may affect the accuracy and interpretability of the results.

9

PDE-based Modeling of Atmospheric Transport and Dispersion of Pollutants

Air pollution transport over industrial regions is governed by a combination of atmospheric flow, turbulent mixing, chemical absorption, and source emission processes. Mathematical models based on advection–diffusion equations provide a tractable framework for studying pollutant concentration dynamics over large spatial scales.

We consider a two-dimensional atmospheric domain consisting of an industrial region and a surrounding urban region. Pollutants are emitted from multiple point sources such as factory smokestacks or power-plant outlets. The pollutant concentration evolves due to:

- horizontal and vertical transport by wind,
- turbulent diffusion,
- atmospheric absorption or chemical decay,
- continuous emission from point sources.

A classical atmospheric diffusion formulation may be found in [Oyjinda et al., 2019](#). We begin with a simplified transport model and then systematically extend it to include more realistic atmospheric effects.

9.1 Basic Atmospheric Transport Model

Let $c(x, y, z, t)$ denote the pollutant concentration at co-ordinate (x, y, z) and time t . We first consider the three-dimensional advection–diffusion equation

$$\frac{\partial c}{\partial t} + u \frac{\partial c}{\partial x} + v \frac{\partial c}{\partial y} + w \frac{\partial c}{\partial z} = k_x \frac{\partial^2 c}{\partial x^2} + k_y \frac{\partial^2 c}{\partial y^2} + k_z \frac{\partial^2 c}{\partial z^2} - \lambda c + S(x, y, z, t). \quad (9.1)$$

For ease of computation we assume that the wind flow in the y direction is not significant and hence we can remove the terms with y and obtain the reduced 2-dimensional advection-diffusion equation

$$\frac{\partial c}{\partial t} + u \frac{\partial c}{\partial x} + w \frac{\partial c}{\partial z} = k_x \frac{\partial^2 c}{\partial x^2} + k_z \frac{\partial^2 c}{\partial z^2} - \lambda c + S(x, z, t). \quad (9.2)$$

The computational domain is $0 \leq x \leq L$, $0 \leq z \leq H$, where, L is the horizontal extent of the region, H is the atmospheric inversion-layer height. The term $\frac{\partial c}{\partial t}$ represents local accumulation or depletion of pollutant concentration. The advection terms $u \frac{\partial c}{\partial x}$, $w \frac{\partial c}{\partial z}$, describe transport due to atmospheric wind velocity. The diffusion terms $k_x \frac{\partial^2 c}{\partial x^2}$, $k_z \frac{\partial^2 c}{\partial z^2}$, model turbulent mixing of pollutants in the horizontal and vertical directions. The term $-\lambda c$ represents pollutant removal due to atmospheric absorption, chemical reactions, or deposition. Finally, $S(x, z, t)$ denotes pollutant emission from external sources.

Suppose there are N point sources located at $(x_1, z_1), \dots, (x_N, z_N)$. A natural representation of the source term is

$$S(x, z, t) = \sum_{k=1}^N Q_k(t) \delta(x - x_k, z - z_k), \quad (9.3)$$

where, $Q_k(t)$ is the emission strength of source k and δ denotes the Dirac delta distribution. In numerical simulations, the delta function is replaced by localized Gaussian approximations or concentrated source cells.

Initial And Boundary Conditions

We assume a cold-start initial condition:

$$c(x, z, 0) = 0. \quad (9.4)$$

The left boundary is treated as a clean inflow boundary:

$$c(0, z, t) = 0. \quad (9.5)$$

At the right boundary we impose a zero-flux condition:

$$\frac{\partial c}{\partial x}(L, z, t) = 0. \quad (9.6)$$

At the ground and upper boundary we impose:

$$\frac{\partial c}{\partial z}(x, 0, t) = 0, \quad \frac{\partial c}{\partial z}(x, H, t) = 0. \quad (9.7)$$

We model reflective boundaries with no flux crossing the domain edges.

9.2 Limitations of the Basic Model and Motivation for Improvement

The atmospheric transport model introduced above captures the essential mechanisms governing pollutant evolution, namely advection, turbulent diffusion, atmospheric absorption, and source emission. However, despite its mathematical simplicity and computational tractability, the model relies on several restrictive assumptions that limit its ability to accurately represent realistic atmospheric transport processes.

The first major simplification is the assumption of constant wind velocity:

$$u = \text{constant}, \quad w = \text{constant}.$$

In realistic atmospheric systems, wind velocity varies significantly with altitude, terrain roughness, temperature gradients, atmospheric stability, and meteorological conditions. In particular, horizontal wind speed generally increases with height due to reduced surface drag, producing atmospheric wind shear. Consequently, constant-coefficient transport models are unable to accurately represent many physically important plume behaviors such as vertical stretching, plume bending, or atmospheric stagnation.

Another major simplification is the assumption of a fixed inversion-layer height:

$$0 \leq z \leq H.$$

In reality, the atmospheric boundary layer evolves continuously over time. During daytime heating, the inversion layer typically rises due to enhanced vertical mixing, while nighttime cooling causes the inversion layer to collapse toward the ground. Since the inversion layer controls the effective vertical mixing region, its dynamics strongly influence near-ground PM_{2.5} concentrations.

Finally, the advection discretization used in the basic FTCS scheme employs central differencing. While mathematically simple, central differencing is known to generate oscillatory and unstable numerical behavior in advection-dominated regimes, particularly when wind velocity varies spatially.

These considerations motivate the development of a more realistic atmospheric transport model incorporating:

1. spatially varying wind velocity,
2. spatially and temporally varying inversion-layer height,
3. improved boundary conditions,
4. more stable numerical discretization schemes.

9.3 Improved Model With Variable Wind and Time-dependent Inversion Height

To obtain a more realistic description of pollutant transport, we replace the constant-velocity model by a variable-coefficient atmospheric transport equation on the time-dependent domain

$$\Omega(t) = \{(x, z) : 0 \leq x \leq L, 0 \leq z \leq H(t)\}.$$

Here, $H(t)$ denotes the inversion-layer height, which changes with time, while the pollutant concentration is denoted by $c(x, z, t)$.

The improved model is

$$\frac{\partial c}{\partial t} + \frac{\partial(uc)}{\partial x} + \frac{\partial(wc)}{\partial z} = \frac{\partial}{\partial x} \left(k_x \frac{\partial c}{\partial x} \right) + \frac{\partial}{\partial z} \left(k_z \frac{\partial c}{\partial z} \right) - \lambda c + S(x, z, t), \quad (9.8)$$

where $u = u(x, z, t)$ and $w = w(x, z, t)$ are the horizontal and vertical wind components, k_x and k_z are the effective turbulent diffusivities, λ is the removal rate, and $S(x, z, t)$ is the emission term.

For multiple point sources, the forcing term may be written as

$$S(x, z, t) = \sum_{m=1}^N Q_m(t) \delta(x - x_m, z - z_m), \quad (9.9)$$

or, in a numerical implementation, by a localized Gaussian approximation of the delta function.

9.3.1 Initial and Boundary Conditions

We use a cold-start initial condition,

$$c(x, z, 0) = 0, \quad 0 \leq x \leq L, 0 \leq z \leq H(0). \quad (9.10)$$

At the left boundary, we impose clean inflow:

$$c(0, z, t) = 0. \quad (9.11)$$

At the right boundary, we impose a zero-flux outflow condition:

$$\frac{\partial c}{\partial x}(L, z, t) = 0. \quad (9.12)$$

At the ground, we retain a no-flux condition for the present model:

$$\frac{\partial c}{\partial z}(x, 0, t) = 0. \quad (9.13)$$

At the inversion layer, the upper boundary is moving in time and is treated as a no-flux cap:

$$\frac{\partial c}{\partial z}(x, H(t), t) = 0. \quad (9.14)$$

9.3.2 Wind-field Parameterization

Only ground-level wind-speed data are assumed to be available. Let $u_g(x, t)$ denote the observed horizontal wind speed near the ground. Since vertical variation is not directly observed, we extend the wind field upward using a prescribed shear profile. A convenient choice is a power-law profile,

$$u(x, z, t) = u_g(x, t) \left(\frac{z + z_0}{z_g + z_0} \right)^\alpha, \quad (9.15)$$

where z_g is the reference height at which the wind data are measured, $z_0 > 0$ is a roughness length, and α controls the vertical shear. This construction ensures that the modeled wind matches the observed ground-level data at $z = z_g$.

For the vertical velocity, one may either prescribe a small background drift or, for a first approximation, set

$$w(x, z, t) \approx 0,$$

since the dominant transport in many plume problems is horizontal. More refined vertical motion can be added later if required.

9.3.3 Inversion-layer Height

The inversion height is modeled as a prescribed time-dependent function. A simple and useful form is

$$H(t) = H_0 + A \sin(\omega t), \quad (9.16)$$

where H_0 is the mean mixing height, A is the oscillation amplitude, and ω is the temporal frequency. This captures the daily rise and fall of the atmospheric boundary layer. If external meteorological data are available, $H(t)$ may also be replaced by an observed or data-assimilated time series.

9.3.4 Numerical Method

We solve (9.8) on a fixed rectangular computational grid

$$x_i = i\Delta x, \quad z_j = j\Delta z, \quad t^n = n\Delta t,$$

with the active domain restricted to the cells satisfying $z_j \leq H(t^n)$ at each timestep.

For the advection terms, we use first-order upwind differencing to maintain stabil-

ity in transport-dominated regimes. For example, if $u_{i,j}^n > 0$, then

$$\frac{\partial c}{\partial x}(x_i, z_j, t^n) \approx \frac{C_{i,j}^n - C_{i-1,j}^n}{\Delta x}, \quad (9.17)$$

while if $u_{i,j}^n < 0$, the forward difference is used instead. The same idea is applied to the vertical advection term involving w .

For the diffusion terms, we use standard second-order central differences:

$$\frac{\partial^2 c}{\partial x^2}(x_i, z_j, t^n) \approx \frac{C_{i+1,j}^n - 2C_{i,j}^n + C_{i-1,j}^n}{\Delta x^2}, \quad (9.18)$$

and similarly in the z -direction.

9.3.5 Semi-implicit Time Integration

To advance the concentration field in time, we use a *semi-implicit discretization*. The main idea is to treat the advection and source terms explicitly, while treating the diffusion terms implicitly. This choice gives a good balance between simplicity, stability, and computational cost.

For the improved model, the governing equation is

$$\frac{\partial c}{\partial t} + \frac{\partial(uc)}{\partial x} + \frac{\partial(wc)}{\partial z} = \frac{\partial}{\partial x} \left(k_x \frac{\partial c}{\partial x} \right) + \frac{\partial}{\partial z} \left(k_z \frac{\partial c}{\partial z} \right) - \lambda c + S(x, z, t).$$

At the discrete level, let

$$C_{i,j}^n \approx c(x_i, z_j, t^n).$$

The semi-implicit method computes the new solution C^{n+1} from the known solution C^n by using an explicit approximation for the transport terms and an implicit approximation for the diffusion terms. Schematically, the update can be written as

$$\frac{C^{n+1} - C^n}{\Delta t} + A(C^n) = D(C^{n+1}) - \lambda C^n + S^n,$$

where $A(C^n)$ denotes the discrete advection operator and $D(C^{n+1})$ denotes the discrete diffusion operator.

Rearranging this relation gives a linear system of the form

$$(I - \Delta t D) C^{n+1} = C^n + \Delta t (A(C^n) - \lambda C^n + S^n),$$

where I is the identity operator. Since the diffusion operator is handled implicitly, the method is significantly more stable than a fully explicit scheme, especially when the diffusion coefficients are non-negligible or when larger time steps are needed.

In practice, the computation at each time step proceeds as follows. First, the wind field $u(x, z, t)$ and $w(x, z, t)$ are evaluated on the grid. Next, the advection terms are approximated using an upwind scheme, because this is more stable for transport-dominated

problems. The diffusion terms are then discretized using second-order central differences. Finally, the resulting sparse linear system is solved to obtain C^{n+1} .

This approach is particularly useful for atmospheric transport problems because it reduces spurious oscillations from advection while preserving the stability advantages of an implicit treatment of diffusion. As a result, the semi-implicit method is well suited for simulating pollutant dispersion under variable wind conditions and a time-dependent inversion layer.

In summary, the improved model uses:

- a variable wind field reconstructed from ground-level wind data using a vertical shear profile,
- a time-dependent inversion height $H(t)$,
- fixed boundary conditions on a moving active domain,
- upwind discretization for advection,
- central discretization for diffusion,
- semi-implicit time stepping for robustness.

10

Validation Using Real Life Data

10.1 Objective of Validation

The purpose of the validation experiment was to investigate whether the proposed PDE-based atmospheric transport model is capable of reproducing the qualitative and quantitative evolution of PM_{2.5} concentration observed in a real industrial region.

Rather than constructing a fully detailed atmospheric CFD model, the objective was to test whether a reduced-order transport-diffusion PDE incorporating:

- wind-driven advection,
- turbulent diffusion,
- time-varying atmospheric boundary layer height,
- and industrial source forcing,

is sufficient to explain the observed temporal variation in PM_{2.5} concentration.

The validation study was carried out using atmospheric and industrial data from the **Durgapur SAIL** industrial region for the date 1 July 2025.

10.2 Study Region and Experimental Setup

The Durgapur SAIL industrial zone was chosen as the validation region because:

- it contains a dominant industrial emission complex,
- the source structure can be approximated by a single effective source,
- meteorological observations were available,
- PM_{2.5} observations were available over multiple timestamps.
- The sensor location used for validation was approximately 8.1 km away from the effective source location.
- Geometrically, the sensor was located approximately: 4° south of the eastward direction from the source. Consequently, the horizontal transport axis of the PDE model was aligned along this direction.

The computational domain was modeled in two dimensions:

$$(x, z),$$

where:

- x denotes the horizontal transport direction,
- z denotes the vertical atmospheric direction.

10.3 Datasets Used

The validation framework incorporated multiple real-world datasets.

PM2.5 Concentration Data

Observed PM2.5 concentrations were collected for the timestamps:

10 AM to 3 PM

on 1 July 2025.

The observed concentrations are listed below:

Hour	Observed PM2.5
10	25.27
11	28.13
12	33.44
13	28.39
14	29.48
15	24.95

These values were used as the primary validation targets for the PDE simulation.

10.3.1 Wind Speed and Wind Direction Data

Meteorological wind observations were available at:

- 10 meter height,
- 100 meter height.

For each timestamp, the dataset contained:

- wind speed at 10 m,
- wind speed at 100 m,
- wind direction at 10 m,
- wind direction at 100 m.

These measurements were used to estimate:

- the effective transport velocity,
- the vertical wind profile,
- the projected transport component along the source-sensor axis.

10.3.2 Boundary Layer Height Data

Boundary layer height (BLH) observations were available for the same date.

The observed values were:

$$H_{\min} = 316 \text{ m}, \quad H_{\max} = 1167 \text{ m}.$$

From these values:

$$H_0 = \frac{H_{\max} + H_{\min}}{2} = 741.5 \text{ m},$$

and

$$A = \frac{H_{\max} - H_{\min}}{2} = 425.5 \text{ m}.$$

A periodic diurnal model was then used:

$$H(t) = 741.5 + 425.5 \sin(0.262t - \pi/2).$$

The angular frequency:

$$\omega = 0.262$$

was chosen based on a 24-hour atmospheric cycle.

10.3.3 Industrial Emission Data

Industrial stack emission reports from Durgapur SAIL provided PM emission strengths for multiple industrial units including:

- coke oven batteries,
- sinter plants,
- blast furnaces.

The reported PM strengths were:

$$41, 35, 38, 40, 112, 81, 9, 13, 10.$$

The total effective source strength was therefore taken as:

$$379.$$

Since absolute PM mass flux was unavailable, the source term was modeled as:

$$Q = \alpha \times 379,$$

where α is an unknown proportionality constant estimated through calibration.

10.4 Modeling Assumptions

The following assumptions were used during validation:

1. The industrial region was approximated as a single effective point source.
2. The transport dynamics were reduced to two spatial dimensions by taking the projection of wind-speed along the source and observer plane.
3. Wind velocity along the y -axis was neglected.
4. Turbulent diffusivities were assumed spatially constant.
5. Wind fields were externally prescribed from observational data.
6. The atmospheric boundary layer acts as a reflective upper boundary.
7. Chemical reactions and secondary aerosol formation were neglected.
8. The initial PM_{2.5} field at 10 AM was assumed spatially uniform.

These assumptions transform the system into a reduced-order atmospheric transport model suitable for numerical simulation.

10.5 Wind Projection and Vertical Wind Modeling

Since the PDE model evolves only along one horizontal transport direction, the observed wind direction must be projected onto the source-sensor axis.

If:

- $\theta(t)$ denotes observed wind direction,
- θ_0 denotes the transport-axis direction,

then the effective velocity becomes:

$$u_{\text{eff}}(t) = u(t) \cos(\theta(t) - \theta_0).$$

The vertical wind profile was estimated using the power-law profile:

$$u(z, t) = u_{10}(t) \left(\frac{z}{10} \right)^{\alpha(t)}.$$

The shear exponent was estimated dynamically:

$$\alpha(t) = \frac{\log(u_{100}(t)/u_{10}(t))}{\log(100/10)}.$$

10.6 Numerical Methodology

The PDE was discretized on a finite difference grid:

$$x_i = i\Delta x, \quad z_j = j\Delta z, \quad t^n = n\Delta t.$$

The numerical scheme used:

- upwind differencing for advection,
- central differencing for diffusion,
- explicit Euler time integration.

The update equation becomes:

$$C_{i,j}^{n+1} = C_{i,j}^n + \Delta t \left(-\text{Advection} + \text{Diffusion} - \lambda C_{i,j}^n + S_{i,j}^n \right).$$

Boundary conditions included:

- inflow condition at the left boundary,
- outflow condition at the right boundary,
- reflective ground condition,
- reflective inversion-layer condition.

10.7 Calibration of the Source Strength

Because the PDE is linear in the source amplitude, the source coefficient α was estimated using least-squares minimization.

Let:

$$\hat{y}(\alpha)$$

denote the predicted PM2.5 vector at the sensor.

Using the linearity of the PDE:

$$\hat{y}(\alpha) = \hat{y}(0) + \alpha (\hat{y}(1) - \hat{y}(0)).$$

The optimal source coefficient is:

$$\alpha^\star = \frac{\delta^\top (y_{\text{obs}} - \hat{y}(0))}{\delta^\top \delta},$$

where:

$$\delta = \hat{y}(1) - \hat{y}(0).$$

This minimizes the mean squared prediction error.

10.8 Simulation Pipeline

At each simulation timestamp:

1. wind data were projected onto the transport axis,
2. the vertical wind profile was estimated,
3. boundary layer height was computed,
4. the PDE was numerically evolved,
5. concentration at the sensor location was extracted,
6. predictions were compared against observed PM2.5.

10.9 Simulation Plots

10.9.1 Hourly PM2.5 Evolution

The numerical solver generated hourly concentration fields showing the evolution of pollutant dispersion over time.

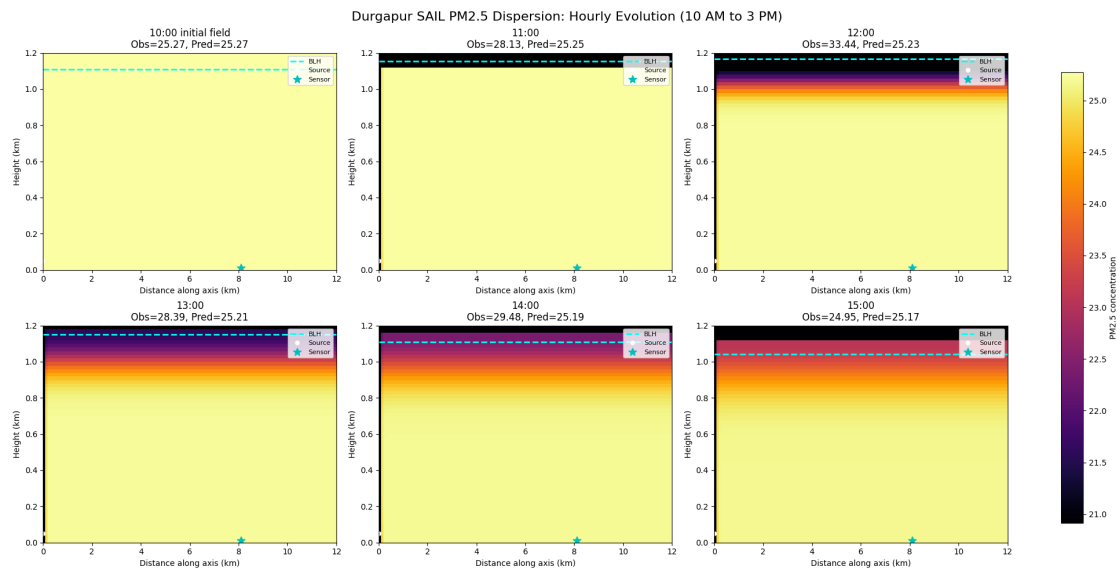


Figure 10.1: Hourly evolution of PM2.5 concentration fields.

10.10 Prediction Results

The observed and predicted PM2.5 values are shown below.

Hour	Observed	Predicted	Error
10	25.27	25.27	0.00
11	28.13	25.25	-2.88
12	33.44	25.23	-8.21
13	28.39	25.21	-3.18
14	29.48	25.19	-4.29
15	24.95	25.17	0.22

The resulting root mean squared error was:

$$\text{RMSE} = 4.1689.$$

10.11 Interpretation of Results

As seen from the plots, as the height of the inversion layer changes, the distribution of concentration of $PM_{2.5}$ also varies. However, the concentrations on the ground level do not seem to vary much. We understand that all the assumptions made in this model are not accurate. We need accurate initial and boundary conditions for the model to provide precise predictions. Since in real life it is a continuous process, setting a certain timestamp at $t = 0$ and assuming uniform concentration is not reasonable. Also we worked with the projected wind-speeds on a certain plain and assumed the influence of wind in the direction normal to that plan is negligible which again might not be the case in real life. Our experiment above can be looked at as a basic implementation of the improved PDE model. The novelty of our improved model lies in incorporating very crucial factors like varying wind-speed and inversion layer height.

10.12 Future Improvements

As discussed earlier The present validation framework remains a reduced-order atmospheric transport model.

Possible future improvements include:

- fully three-dimensional transport,
- Treating the source of emission as n -point sources instead of a single point source.
- Incorporating spatially varying diffusivities.
- implicit or semi-implicit numerical methods.

Nevertheless, the framework demonstrates how PDE-based atmospheric transport models can be connected to real observational datasets for physically interpretable $PM_{2.5}$ analysis.

11

Conclusion

We first carried out a multivariate time-series analysis using Vector Autoregressive (VAR) models and Granger causality analysis on atmospheric and pollutant variables. The VAR framework enabled us to study the temporal interdependence between meteorological factors and pollutant concentrations, while Granger causality diagrams provided a graphical representation of directional predictive relationships among the variables. The results revealed that the interactions between temperature and particulate pollutants are highly location-dependent, with both bidirectional and unidirectional causal structures observed across different stations. These statistical findings provide insight into the dynamic behaviour of atmospheric variables and motivate the need for more detailed physical modelling of pollutant transport.

Motivated by these observations, we developed a two-dimensional atmospheric transport model for pollutant dispersion from multiple point sources. Starting from a classical advection–diffusion equation, we introduced spatially varying wind fields and a time-dependent inversion layer to obtain a more realistic atmospheric model.

The resulting PDE framework remains computationally tractable while incorporating several physically important atmospheric effects. The model can be numerically approximated using finite-difference methods and validated against real atmospheric monitoring data.

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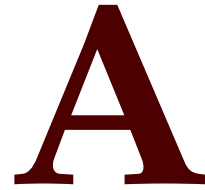
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Appendices



Granger Causality Diagrams

Diagram Link	Location	Observations
View	WBPCB Office, Salt Lake, North 24 Parganas	External temperature Granger causes PM _{2.5} and PM ₁₀ . This means that past values of temperature significantly improve the prediction accuracy of PM _{2.5} and PM ₁₀ concentrations.
View	Vidyasagar University, Paschim Medinipur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-5, Sidho-Kano-Birsha University, Purulia	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-13, Sector V Industrial Township Authority, Nabadiganta Bhaban, North 24 Pgs.	Temperature Granger-causes SO ₂ and NO ₂ , which in turn Granger-cause PM ₁₀ and PM _{2.5} . There is no direct causal relationship between temperature and particulate matter concentrations.
View	DXAQMS-7, West Bengal State University, North 24 Parganas	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-8, Bandel Thermal Power Station, WBPCDL, Hooghly	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-2, Jhargram Raj College, Jhargram	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-9, Paschim Bardhaman	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-10, Purba Bardhaman	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀

Diagram Link	Location	Observations
View	DXAQMS-11, Kolkata	PM ₁₀ and PM _{2.5} both Granger causes temperature. Temperature Granger-causes SO ₂ which Granger causes PM ₁₀ and PM _{2.5}
View	DXAQMS-6, Nadia	Bidirectional causality between temperature and PM ₁₀ , and temperature Granger causes PM _{2.5} .
View	DXAQMS-14, Purba Medinipur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-12, Burdwan University, Purba Bardhaman	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-19, Diamond Harbour, South 24 Pgs	Temperature Granger causes PM ₁₀ and PM _{2.5}
View	DXAQMS-16, Malda	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-22, Darjeeling	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-21, Darjeeling	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-24, Paschim Bardhaman	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-23, Paschim Bardhaman	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-17, Uttar Dinajpur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-20, Coochbehar	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-18, Dakshin Dinajpur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-26, Kalimpong	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-28, Paschim Bardhaman	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-72, North 24 Pgs	Temperature Granger causes PM ₁₀ and PM _{2.5} .
View	DXAQMS-29, Darjeeling	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-32, Birbhum	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-30, Alipurduar	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-31, Murshidabad	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀

Diagram Link	Location	Observations
View	DXAQMS-27, Jalpaiguri	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-33, Purba Medinipur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-35, Bankura Christian College, Bankura	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-37, Hooghly	Temperature Granger causes PM _{2.5} and PM ₁₀
View	DXAQMS-38, Kolkata	Temperature Granger causes NO ₂ which Granger causes PM _{2.5} , and PM ₁₀
View	DXAQMS-40, Bankura	Bidirectional Granger causality between temperature and PM ₁₀ , and PM _{2.5} Granger causes temperature.
View	DXAQMS-42, Birbhum	PM _{2.5} and PM ₁₀ Granger cause temperature. Temperature Granger causes NO ₂ which Granger causes PM ₁₀ and PM _{2.5}
View	DXAQMS-71, North 24 Pgs	Bidirectional causality between temperature and PM _{2.5} , and temperature Granger causes PM ₁₀
View	DXAQMS-41, North 24 Pgs	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-43, North 24 Pgs	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-39, Murshidabad	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-44, Murshidabad	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-46, Purba Bardhaman	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-48, Bankura	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-49, Murshidabad	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-50, Kolkata	Strong evidence of temperature Granger causing PM ₁₀ , and weaker evidence of Granger causing PM _{2.5}
View	DXAQMS-51, Paschim Bardhaman	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-52, Hooghly	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-53, North 24 Pgs	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀

Diagram Link	Location	Observations
View	DXAQMS-55, Kolkata	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀ , however evidence against null hypothesis is weaker (p-value between 0.05 and 0.1) for testing PM _{2.5} Granger causing temperature.
View	DXAQMS-54, Purulia	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-56, Purulia	Temperature Granger causes PM _{2.5} , and PM ₁₀
View	DXAQMS-47, Howrah	Temperature Granger causes PM _{2.5} , and PM ₁₀
View	DXAQMS-57, North 24 Pgs.	Temperature Granger causes PM _{2.5} , and PM ₁₀
View	DXAQMS-58, North 24 Pgs.	Temperature Granger causes PM _{2.5} , and PM ₁₀
View	DXAQMS-59, Purulia	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-60, Purulia	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-61, Purulia	Temperature Granger causes PM _{2.5} , and PM ₁₀
View	DXAQMS-62, Bankura	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-63, Bankura	PM _{2.5} Granger causes temperature, and PM ₁₀ also Granger causes temperature. Temperature Granger causes SO ₂ which Granger causes PM _{2.5} and PM ₁₀ .
View	DXAQMS-64, South 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-67, South 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-65, Nadia	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-66, Murshidabad	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-68, Birbhum	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-69, South 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-70, South 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀ . However, evidence against null hypothesis is weaker (p-value between 0.05 and 0.1) for testing temperature Granger causing PM _{2.5} .
View	DXAQMS-73, Murshidabad	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀

Diagram Link	Location	Observations
View	DXAQMS-78, Bankura	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-75, Darjeeling	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-74, Jalpaiguri	Temperature Granger causes PM ₁₀ and PM _{2.5}
View	DXAQMS-76, Alipurduar	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-77, Coochbehar	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-79, Darjeeling	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-80, Malda	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-81, Dakshin Dinajpur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-82, Dakshin Dinajpur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-83, Uttar Dinajpur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-84, South 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-85, South 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-86, Howrah	Temperature Granger causes PM _{2.5}
View	DXAQMS-87, Kolkata	Temperature does not Granger cause PM _{2.5} and temperature does not Granger cause PM ₁₀
View	DXAQMS-88, Purba Medinipur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-89, Purba Medinipur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-90, Kolkata	PM ₁₀ Granger causes temperature and PM _{2.5} Granger causes temperature
View	DXAQMS-93, Kolkata	Temperature Granger causes PM _{2.5} , and temperature also Granger causes PM ₁₀ . PM ₁₀ Granger causes temperature, but the p-value for the test is between 0.05 and 0.10.
View	DXAQMS-94, North 24 Pgs.	Temperature Granger causes PM _{2.5} , and temperature also Granger causes PM ₁₀ . PM ₁₀ Granger causes temperature, but the p-value for the test is between 0.05 and 0.10.
View	DXAQMS-95, North 24 Pgs.	Temperature Granger causes PM _{2.5} , and temperature Granger causes PM ₁₀

Diagram Link	Location	Observations
View	DXAQMS-96, Coochbehar	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-97, North 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀ . However, the Granger causality between PM ₁₀ and temperature and PM _{2.5} and temperature, is not significant at the 5% level, it is significant at the 10% level.
View	DXAQMS-98, Kolkata	PM _{2.5} Granger causes temperature and PM ₁₀ Granger causes temperature. Temperature Granger causes PM ₁₀ at the 10% level of significance.
View	DXAQMS-99, Uttar Dinajpur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-100, Jalpaiguri	Temperature Granger causes PM _{2.5} at 5% level of significance. PM _{2.5} Granger causes temperature at 10% level of significance. Temperature Granger causes PM ₁₀ at 10% level of significance, and PM ₁₀ Granger causes temperature at 10% level of significance.
View	DXAQMS-101, North 24 Pgs.	No significant Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-104, Purba Medinipur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-105, South 24 Pgs.	Temperature Granger causes PM ₁₀ and PM _{2.5} at 5% level of significance. PM ₁₀ Granger causes temperature at 10% level of significance.
View	DXAQMS-106, Purba Medinipur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-107, Purba Medinipur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-108, Hooghly	Bidirectional causality between temperature and PM _{2.5} , and temperature Granger causes PM ₁₀ at 5% level of significance. PM ₁₀ Granger causes temperature at 10% level of significance.
View	DXAQMS-109, Hooghly	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-110, North 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-111, Purba Medinipur	Bidirectional causality between temperature and and PM ₁₀ , PM _{2.5} Granger causes temperature at 5% level of significance
View	DXAQMS-112, Purba Medinipur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀

Diagram Link	Location	Observations
View	DXAQMS-114, North 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-115, Purba Medinipur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-116, North 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-117, Alipurduar	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-118, Paschim Bardhaman	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-119, North 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-120, Paschim Bardhaman	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-121, Paschim Bardhaman	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-122, Paschim Bardhaman	Temperature Granger causes PM _{2.5} , and temperature Granger causes PM ₁₀
View	DXAQMS-123, Howrah	No significant Granger causality between temperature and PM _{2.5} , or between temperature and PM ₁₀
View	DXAQMS-124, Paschim Bardhaman	Temperature Granger causes PM _{2.5} , and temperature Granger causes PM ₁₀
View	DXAQMS-126, Paschim Medinipur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-127, Paschim Medinipur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-128, Paschim Bardhaman	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-129, Paschim Medinipur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-130, Howrah	Temperature Granger causes PM ₁₀ at 10% level of significance
View	DXAQMS-159, North 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} . Temperature Granger causes PM ₁₀ at 5% level of significance. PM ₁₀ Granger causes temperature at 10% level of significance.
View	DXAQMS-131, Kolkata	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-132, Paschim Medinipur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-133, Paschim Medinipur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀

Diagram Link	Location	Observations
View	DXAQMS-134, Paschim Medinipur	PM ₁₀ Granger causes temperature at 10% level of significance
View	DXAQMS-135, Howrah	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-136, Uttar Dinajpur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-137, Uttar Dinajpur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-138, North 24 Pgs.	Temperature Granger causes PM _{2.5} , and PM ₁₀ at 5% level of significance
View	DXAQMS-139, Kolkata	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-140, Paschim Bardhaman	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-141, North 24 Pgs.	Temperature Granger causes PM _{2.5} , and PM ₁₀ at 5% level of significance
View	DXAQMS-142, Kolkata	PM _{2.5} Granger causes temperature, and PM ₁₀ Granger causes temperature
View	DXAQMS-143, North 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-144, North 24 Pgs.	Temperature Granger causes PM _{2.5} , and PM ₁₀
View	DXAQMS-145, North 24 Pgs.	Bidirectional causality between temperature and and PM ₁₀ . Temperature Granger causes PM _{2.5}
View	DXAQMS-146, Jhargram	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-148, North 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-149, Uttar Dinajpur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-150, Murshidabad	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-151, Paschim Bardhaman	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-152, Nadia	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-153, Kolkata	Temperature Granger causes PM _{2.5} , and PM ₁₀
View	DXAQMS-154, Paschim Bardhaman	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-155, Paschim Bardhaman	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀

Diagram Link	Location	Observations
View	DXAQMS-157, Paschim Medinipur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-158, Kolkata	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-163, Murshidabad	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-165, North 24 Pgs.	No significant Granger causality between temperature and PM _{2.5} , or between temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-166, Kolkata	Temperature Granger causes PM _{2.5} , and PM ₁₀ at 5% level of significance.
View	DXAQMS-168, North 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-169, Nadia	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-170, North 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-171, Kolkata	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-172, North 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-173, Purba Bardhaman	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-174, Birbhum	Temperature Granger causes PM _{2.5} , and PM ₁₀
View	DXAQMS-175, Purba Bardhaman	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-176, North 24 Pgs.	Temperature Granger causes PM _{2.5} , and PM ₁₀
View	DXAQMS-177, South 24 Pgs.	PM _{2.5} Granger causes temperature and PM ₁₀ Granger causes temperature
View	DXAQMS-178, Hooghly	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-179, North 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-180, Darjeeling	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-182, Coochbehar	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-183, Coochbehar	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-184, North 24 Pgs.	Both PM _{2.5} and PM ₁₀ Granger cause temperature.

Diagram Link	Location	Observations
View	DXAQMS-185, Alipurduar	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-186, Alipurduar	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-188, Hooghly	Temperature Granger causes both PM _{2.5} , and PM ₁₀
View	DXAQMS-189, South 24 Pgs.	Temperature Granger causes PM _{2.5} , and PM ₁₀ . PM _{2.5} Granger causes temperature at the 10% level of significance.
View	DXAQMS-190, Hooghly	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-193, North 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-195, North 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-196, South 24 Pgs.	Temperature Granger causes PM _{2.5} , and PM ₁₀
View	DXAQMS-197, Hooghly	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-198, North 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-199, Birbhum	PM _{2.5} , and PM ₁₀ both Granger cause temperature. Temperature Granger causes both PM ₁₀ and PM _{2.5} at 10% level of significance.
View	DXAQMS-200, Paschim Medinipur	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-201, Paschim Medinipur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-202, Paschim Medinipur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-203, Howrah	Bidirectional Granger causality between temperature and PM ₁₀ . Temperature Granger causes PM _{2.5} at 5% level of significance whereas PM _{2.5} Granger causes temperature at 10% level of significance.
View	DXAQMS-204, Dakshin Dinajpur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-205, Uttar Dinajpur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-207, Nadia	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-208, Paschim Medinipur	PM _{2.5} and PM ₁₀ Granger cause temperature. Temperature Granger causes PM ₁₀ at the 10% level of significance.

Diagram Link	Location	Observations
View	DXAQMS-209, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-210, Nadia	PM _{2.5} and PM ₁₀ Granger cause temperature at 10% level of significance.
View	DXAQMS-211, Nadia	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-212, North 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-213, South 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-214, Purba Medinipur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-215, Purba Medinipur	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-216, Hooghly	PM _{2.5} as well as PM ₁₀ Granger cause temperature. Temperature Granger causes PM _{2.5} at 10% level of significance.
View	DXAQMS-217, South 24 Pgs.	PM _{2.5} as well as PM ₁₀ Granger cause temperature.
View	DXAQMS-218, Hooghly	Temperature Granger causes both PM _{2.5} as well as PM ₁₀
View	DXAQMS-219, Hooghly	Temperature Granger causes both PM _{2.5} as well as PM ₁₀
View	DXAQMS-221, Nadia	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-222, Nadia	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-223, Howrah	Temperature Granger causes PM ₁₀ at 5% level of significance. Temperature Granger causes PM _{2.5} at 10% level of significance.
View	DXAQMS-224, North 24 Pgs.	Bidirectional causality between temperature and PM ₁₀ at 10% level of significance. PM _{2.5} Granger causes temperature at 10% level of significance.
View	DXAQMS-225, Uttar Dinajpur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-226, Uttar Dinajpur	Temperature Granger causes PM _{2.5} as well as PM ₁₀ at 5% level of significance.
View	DXAQMS-227, Uttar Dinajpur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-228, Uttar Dinajpur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀

Diagram Link	Location	Observations
View	DXAQMS-229, Uttar Dinajpur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-230, Uttar Dinajpur	PM ₁₀ as well as PM _{2.5} Granger cause temperature at the 10% level of significance
View	DXAQMS-231, Uttar Dinajpur	Temperature Granger causes PM _{2.5} , as well as PM ₁₀ at 5% level of significance
View	DXAQMS-232, Uttar Dinajpur	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-233, Kolkata	Temperature Granger causes PM _{2.5} , as well as PM ₁₀ at 5% level of significance.
View	DXAQMS-234, Nadia	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-235, Purba Bardhaman	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-237, North 24 Pgs.	Temperature Granger causes PM ₁₀ and PM _{2.5} at 5% level of significance
View	DXAQMS-238, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-239, South 24 Pgs	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-240, Purulia	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-241, Bankura	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-242, Murshidabad	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-243, North 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-244, South 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-245, Bankura	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-247, Paschim Medinipur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-248, Malda	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-249, Uttar Dinajpur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-250, Purba Bardhaman	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-251, Purba Bardhaman	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀

Diagram Link	Location	Observations
View	DXAQMS-252, Purba Bardhaman	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-253, Murshidabad	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-254, Jhargram	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-257, Dakshin Dinajpur	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-260, Darjeeling	Temperature Granger causes PM ₁₀ and PM _{2.5} . PM ₁₀ as well as PM _{2.5} Granger cause temperature at the 10% level of significance.
View	DXAQMS-261, North 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-262, Hooghly	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-263, Birbhum	PM ₁₀ as well as PM _{2.5} Granger cause temperature, and temperature Granger causes PM ₁₀ at the 10% level of significance.
View	DXAQMS-264, Birbhum	Temperature Granger causes PM _{2.5} , and PM ₁₀ . PM _{2.5} Granger causes temperature at 10% level of significance.
View	DXAQMS-265, Birbhum	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-266, Murshidabad	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-268, Kolkata	PM ₁₀ Granger causes temperature at 10% level of significance.
View	DXAQMS-269, Nadia	PM ₁₀ as well as PM _{2.5} Granger cause temperature, and temperature Granger causes PM ₁₀ at the 10% level of significance.
View	DXAQMS-271, Coochbehar	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-272, Purba Medinipur	Temperature Granger causes PM _{2.5} , and temperature Granger causes PM ₁₀ at 5% level of significance.
View	DXAQMS-273, South 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} , and temperature and PM ₁₀
View	DXAQMS-275, North 24 Pgs.	PM _{2.5} , as well as PM ₁₀ Granger cause temperature at 5% level of significance.
View	DXAQMS-276, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.

Diagram Link	Location	Observations
View	DXAQMS-277, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-278, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-279, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-280, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-281, North 24 Pgs.	Temperature Granger causes PM _{2.5} , and temperature Granger causes PM ₁₀ at 5% level of significance. PM _{2.5} Granger causes temperature at 10% level of significance.
View	DXAQMS-282, North 24 Pgs.	Temperature Granger causes PM _{2.5} , as well as PM ₁₀ at the 5% level of significance.
View	DXAQMS-283, North 24 Pgs.	Temperature Granger causes PM _{2.5} , as well as PM ₁₀ at the 5% level of significance. PM ₁₀ Granger causes temperature at 10% level of significance.
View	DXAQMS-284, North 24 Pgs.	Temperature Granger causes PM _{2.5} , as well as PM ₁₀ at the 5% level of significance.
View	DXAQMS-285, North 24 Pgs	Temperature Granger causes PM _{2.5} , as well as PM ₁₀ at the 5% level of significance. PM ₁₀ Granger causes temperature at 5% level of significance and PM _{2.5} Granger causes temperature at 10% level of significance.
View	DXAQMS-286, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-287, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-288, Howrah	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-289, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-290, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.

Diagram Link	Location	Observations
View	DXAQMS-291, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-292, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-293, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-294, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-295, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-296, North 24 Pgs.	Bidirectional causality between temperature and PM _{2.5} at 5% level of significance. Temperature Granger causes PM ₁₀ at 5% level of significance. PM ₁₀ Granger causes temperature at 10% level of significance.
View	DXAQMS-297, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-298, North 24 Pgs.	Bidirectional Granger causality between temperature and PM ₁₀ at 5% level of significance. Temperature Granger causes PM _{2.5} at 5% level of significance.
View	DXAQMS-299, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-300, North 24 Pgs.	Temperature Granger causes PM ₁₀ at 5% level of significance, and PM _{2.5} at 5% level of significance.
View	DXAQMS-302, North 24 Pgs.	Bidirectional Granger causality between temperature and PM ₁₀ at 5% level of significance. Temperature Granger causes PM _{2.5} at 5% level of significance.
View	DXAQMS-303, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-304, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.

Diagram Link	Location	Observations
View	DXAQMS-306, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-307, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-308, Howrah	Bidirectional Granger causality between temperature and PM _{2.5} at 5% level of significance. Temperature Granger causes PM ₁₀ at 5% level of significance.
View	DXAQMS-309, Howrah	Temperature Granger causes PM _{2.5} , as well as PM ₁₀ at 5% level of significance. PM _{2.5} and PM ₁₀ Granger cause temperature at 10% level of significance.
View	DXAQMS-310, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-311, Hooghly	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-312, Howrah	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-313, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-314, Howrah	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-315, Hooghly	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-316, Hooghly	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-317, Hooghly	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-318, Hooghly	No significant Granger causality between temperature and PM _{2.5} , or between temperature and PM ₁₀ .
View	DXAQMS-319, Hooghly	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.

Diagram Link	Location	Observations
View	DXAQMS-320, Howrah	PM _{2.5} , as well as PM ₁₀ Granger cause temperature at 5% level of significance.
View	DXAQMS-321, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-322, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-323, Howrah	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-324, Howrah	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-325, North 24 Pgs.	Temperature Granger causes PM _{2.5} , as well as PM ₁₀ at 5% level of significance.
View	DXAQMS-326, Hooghly	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-327, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-328, Howrah	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-329, North 24 Parganas	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-331, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-330, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-332, Howrah	Temperature Granger causes PM ₁₀ at 5% level of significance. PM ₁₀ Granger causes temperature at 10% level of significance. Temperature Granger causes PM _{2.5} at 10% level of significance, and PM _{2.5} Granger causes temperature at 10% level of significance.
View	DXAQMS-333, Hooghly	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.

Diagram Link	Location	Observations
View	DXAQMS-334, North 24 Pgs.	Temperature Granger causes PM ₁₀ as well as PM _{2.5} at 5% level of significance.
View	DXAQMS-335, Howrah	No Granger causality between temperature and PM _{2.5} , or between temperature and PM ₁₀
View	DXAQMS-336, Howrah	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-337, Howrah	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-339, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-340, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-341, North 24 Pgs.	Temperature Granger causes both PM ₁₀ and PM _{2.5} at 5% level of significance. PM ₁₀ Granger causes temperature at 5% level of significance.
View	DXAQMS-342, North 24 Pgs.	Bidirectional Granger causality between temperature and PM ₁₀ at 5% level of significance. Bidirectional Granger causality between PM ₁₀ and PM _{2.5} at 5% level of significance.
View	DXAQMS-343, North 24 Pgs.	No Granger causality between temperature and PM _{2.5} , or between temperature and PM ₁₀ .
View	DXAQMS-344, Howrah	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.
View	DXAQMS-345, North 24 Pgs.	Bidirectional Granger causality between temperature and PM _{2.5} , and temperature and PM ₁₀ at 5% level of significance.

